

Ribo-seq signal model: data, training, evaluation

one-hot attention architecture (orf_v2_attn), union universe

A

Data construction and leave-one-tissue-out split



7 tissues -> TRAIN

1 tissue -> HELD OUT (never seen)

Ribo-seq	per-nt P-site counts = the LABEL
RNA-seq	per-nt coverage = an INPUT
GENCODE v49	sequence + reading-frame track = INPUTS

84,472 transcripts
protein-coding + lncRNA, up to 10 kb, chrM excluded

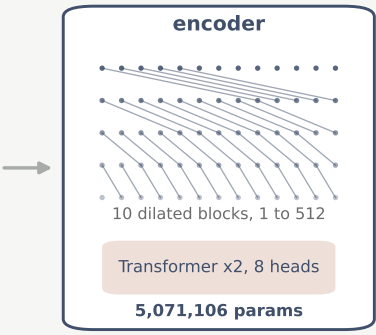
validation genes are chromosome-disjoint from training
so early stopping cannot leak into the held-out result

B

Training pipeline

10-track per-nucleotide encoding

one-hot A C G U	4
reading frame + start/stop	5
log RNA-seq coverage	1



profile head
softmax over L

count head
scalar log total

objective
multinomial NLL (profile)
+ 0.1 x MSE (log count)

select on median per-transcript
Pearson (validation)
best epoch 21 | 14.2 h | 1x A5500



C

Evaluation: four axes of generalization

In-distribution held-out tissue profile $r = 0.659$ CDS F1 = 0.953	Cross-study Ruiz-Orera human CDS F1 = 0.931 beats obs ceiling	Cross-species 5 mouse datasets CDS F1 0.960 to 0.995 human-trained only	Proteogenomics 12 macrophage pops 440 novel peptides 1% class FDR
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Trained once on human tissue Ribo-seq; every axis above is data the model never saw during training.
No ribosome profiling is required at inference: the model predicts the profile from sequence + RNA-seq alone.